

Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose

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Abstract

Deployed LLM agents are organisations of hand-offs: planner to worker to reviewer to aggregator. Every hand-off is a trust boundary, and a prompt injection that hijacks one agent propagates into the next agent’s input. Existing benchmarks measure one agent over one hop; they cannot quantify whether a compromise survives transfer, how far it travels, or where to intervene. We present *Contagion*, a framework for measuring propagation in LLM agent networks using two independent protocols: a *controlled per-edge* protocol that estimates conditional survival $s_i = \Pr[C_i=1 \mid C_{i-1}=1]$, and a *natural-run* protocol that measures end-to-end ASR and the reproduction number R_0 . Four results follow. (i) $ASR = \prod_i s_i^{\text{nat}}$ is an algebraic identity; the assumption worth testing is whether isolated per-hop estimates *transport* to the in-chain regime—they can underestimate by $3.75\times$ ($p = 0.0002$) or transport within noise ($p = 0.93$), and the depth-error slope distinguishes the two cases. (ii) The mechanism is payload *form*: the same target survives at 0.067 arriving as a bare assertion and at 0.967 embedded in a work product. (iii) The epidemic threshold view: on strictly acyclic (feed-forward) graphs the next-generation matrix is nilpotent in the standard linearized form ($\rho(M) = 0 < 1$) as a structural property of acyclicity rather than an empirical safety guarantee. Accordingly, we treat R_0 as a descriptive, finite-horizon reproduction statistic (with its definition, denominator, generation convention, and scope stated in the theory) and report $R_0 < 1$ alongside ASR of 0.45–0.73. (iv) Defence efficacy is unidentifiable without a no-defence baseline: the same redaction code is essential or irrelevant depending solely on the model’s unprotected susceptibility.

CCS Concepts

• **Computing methodologies** → Multiagent systems; Natural language processing; • **Security and privacy** → Social engineering.

Keywords

Multiagent systems, Prompt injection, Security measurement, Epidemic threshold, LLM agents, Benchmarks

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1 Introduction

LLM agents are deployed as *organisations*, not as singletons: a planner decomposes a task, workers execute subtasks, a reviewer checks output, an aggregator merges results. Frameworks such as MetaGPT [18], AutoGen [35] and AgentVerse [10] route free text between these agents, and tool responses or retrieved documents flow directly into prompts. Every hand-off is a *trust boundary*: the receiving agent must consume text produced by another agent, and that text may have been shaped by content the upstream agent read from an untrusted channel.

Indirect prompt injection exploits this. An attacker who controls any text an agent reads can embed an instruction; if the agent follows it, its *output* now carries the attacker’s intent, and that output is the *input* of the next agent, so the injection does not stay where it landed. Single-agent work establishes that one agent can be hijacked through a tool response [12, 38, 39] and that an injection can persist within one application [16]; that it also propagates agent-to-agent is now established as well [3, 24]. What has not been established is a framework answering the question an engineer actually faces: *if one agent is injected, how far does the injection travel, and what determines whether it survives each hand-off?*

Why the standard summary statistic is not enough. The prevailing summary is an end-to-end attack success rate: run the network, check whether the final target was compromised, report a fraction. This number is real but lossy in two ways that we demonstrate empirically rather than assume.

(a) *It hides where failure happens.* In a five-agent chain per-hop survival ranges from 0.133 (worker) to 0.867 (reviewer)—a $6.5\times$ spread within one model and attack—so “the attack failed” and “the attack failed at one specific edge” are findings only the per-hop view separates.

(b) *It is not comparable across models without a no-defence baseline.* A resistant frontier model is compromised at 0.10 with *no defence at all*, so a defence driving $0.10 \rightarrow 0.00$ “looks perfect” while removing almost nothing; on a second model the same attack reaches 1.00 and the same defence drives it to 0.00—same defence, attack and code, opposite conclusions. We call this the *floor effect* and treat it as a first-class threat to validity.

What we build. The framework rests on three commitments: per-hop survival s_i from a *controlled per-edge* protocol (force the source compromised, feed its output to the receiver, judge the receiver) and end-to-end ASR from an *independent natural-run* protocol; an injected task semantically competitive with the agent’s legitimate assignment rather than a marker echo, which saturates at 1.0; and reported resolution—Wilson intervals [6, 34], a bootstrap test [14] with a stated minimum detectable effect, and estimators validated on synthetic ground truth. What distinguishes this from another injection benchmark is that its central results are structural:

composing per-hop rates in a chain is an *identity* plus a transportability hypothesis, and the epidemic threshold $\rho(M) < 1$ [20, 28] is *identically* satisfied on any feed-forward graph and so certifies nothing—properties of the graph and its roles, undefined for a single agent.

Findings, stated as prescriptions. We apply the framework to five models from five independent developers—coverage differs by claim (cross-model susceptibility spans all five, the depth profile three, the topology comparison three models with two instances for Llama)—and reach four conclusions, each of which changes what a practitioner should do.

- (1) **Do not compose isolated injection results into pipeline predictions without first checking that they transport.** Where the isolated rate fails to transport, its error *grows with distance* (+11.8 points per hop, rank correlation +1.00); on a model that does transport the slope is an order of magnitude shallower—so the depth profile’s slope is itself the diagnostic (Section 5.7).
- (2) **Report per-hop survival conditioned on the receiving role, together with ASR and legitimate-task retention, and always include the no-defence baseline.** Per-hop survival varies $6.5\times$ across receiver roles within one model and attack (Section 5), and a defence’s efficacy is otherwise unidentifiable (the floor effect, Section 5.5).
- (3) **Do not use $R_0 < 1$ as a safety criterion for a feed-forward agent network.** The spectral threshold is identically satisfied on any acyclic graph, and we measure $R_0 < 1$ alongside end-to-end success of 0.45–0.73 (Section 3.8). Reach and reproduction are distinct risks.
- (4) **Treat defence placement as a structural question.** Only nodes on an entry–target path matter, and among those all are equally effective, so placement is decided by the graph and by instrumentation cost—not by measured per-hop survival (Section 3.8).

We also report one of our own initially significant results that proved to be an artefact of the controlled protocol, and the change that removed it (Section 5.3); in a measurement study the protocol is part of the result.

2 Related Work and Positioning

Single-agent indirect injection. Prompt injection as an attack class was first named and systematised by Pérez and Ribeiro [29], who showed that an attacker controlling any text an LLM reads can override prior instructions. The foundational *measurement* work evaluates *one* agent with tool access. InjecAgent [38] builds 1,054 test cases across 30 LLMs and reports an attack success rate of 24% for a ReAct [36] GPT-4 agent, with the *content-freedom* of the injected field dominating vulnerability. AgentDojo [12] provides a dynamic evaluation environment for one agent over one hop, and Agent Security Bench [39] broadens this to ten application scenarios with an average attack success rate above 84%. Tool calls are prompted or learned [30, 31]; agent benchmarks [4, 19, 25] report the same scalar. Liu and Gong [26] formalise injection as $\tilde{x} = \mathcal{A}(x_{\text{target}}, s_{\text{inj}}, x_{\text{inj}})$ and introduce the ASV/MR metrics we reuse. All of these terminate at one hop: they cannot express whether a compromise survives transfer. Liu and Gong also report that attack

success correlates *positively* with model size—an early signal, which our cross-model results sharpen, that susceptibility is not a simple function of capability.

Propagation across agents, and the closest prior work. Greshake et al. [16] demonstrate a self-replicating injection “worm” within a single application, establishing persistence but not topology-dependent spread. Prompt Infection [24] lifts the virus metaphor to agent-to-agent spread and shows that compromised agents infect one another even under local messaging; it proposes tagging agent output to restore the user/agent provenance boundary. The nearest work is ACIArena [3], which names the threat *agent cascading injection* and benchmarks 1,356 cases over six multi-agent frameworks and three attack surfaces. MAST [7] separately catalogues multi-agent failures and attributes 32.3% of them to inter-agent misalignment, and concurrent work reports aggregate compromise rates through hierarchical chains [1].

ACIArena and Prompt Infection report *outcome* rates per attack surface or per application; we report the *conditional* probability $\Pr[C_v=1 \mid C_u=1]$ for each edge, which makes failure localisable, and test whether an isolated rate predicts its in-chain rate—a question an aggregate rate cannot pose. ACIArena also reports that narrowly scoped defences fail to transfer; we make the mechanism explicit as the *floor effect* (Section 5.5), fixed by a matched no-defence baseline.

Defences. Adaptive attacks against single-agent injection defences break eight defences at $> 50\%$ ASR, including paraphrasing, using token-level optimisation [37]; our negative result on semantic paraphrasing inside a chain is consistent with that finding. Structured queries [9], encoding [17], and removing the agent’s authority to act on untrusted text [11] instead constrain how such content reaches the model. Our attacker instead uses *natural-language structural obfuscation* (splitting or spacing a literal target), not white-box token optimisation.

Contagion and thresholds in networks. Treating propagation as contagion over a graph is classical, beginning with the SIR formulation [22]. Threshold models make an individual’s adoption a function of how many neighbours have already adopted [15, 32]; epidemic models supply the complementary criterion that spread is subcritical when the spectral radius of the next-generation matrix falls below one, a threshold that can vanish on heavy-tailed topologies [20, 27, 28]; small-world structure [33] makes cascades tractable, and robustness to node failure follows from the error-tolerance literature [2]. Controlled experiments show that *complex* contagions—requiring several concurrent exposures—spread more slowly and reach fewer nodes [8]: the network-level reason to expect per-edge survival, not per-agent susceptibility, to govern reach. Seeding nodes to maximise reachability is NP-hard in general [21] but linear-time on acyclic graphs by topological traversal, which is why Eq. (3) is exact on our networks. This is the lens recent theory work on multiplex Markov contagion in agent swarms adopts [23]. Our results sharpen where the analogy fails for LLM agent networks: for strictly acyclic feed-forward graphs the standard linearized next-generation matrix is nilpotent ($\rho(M) = 0 < 1$) as a structural property of acyclicity rather than an empirical safety guarantee (Section 3.8); the empirical reproduction number can be below one while reachability stays high; and the per-hop rates such

an analysis consumes do not transport from isolated measurement (the transportability depends on graph structure and role/context as discussed in Section ??).

3 Framework: Measuring Propagation

3.1 Network and threat model

An agent network is a directed graph $G = (V, E)$. Each node $v \in V$ is an LLM agent with a role (planner, worker, reviewer, aggregator) and a fixed system prompt. Each edge $(u, v) \in E$ is a *trust boundary*: text produced by u is consumed by v as untrusted content. We instantiate chain, star and balanced binary tree topologies.

The attacker controls the text of one untrusted channel reaching the *entry agent* v_0 , following InjecAgent’s indirect channel [38]. We adopt the standard construction that the entry is compromised by definition, $C_0 = 1$: the measurement targets *propagation*, not entry-point susceptibility. Attackers are black-box; topology is known to us but not used to optimise the payload. Every agent also receives a *legitimate assignment* from a trusted channel (a short operational task), so that an agent which resists injection merely by refusing to work is not credited as a defence. Injected content is therefore framed as benign-looking infrastructure text rather than as an overt command; in a pilot this framing was the difference between 0% and 60–100% compliance.

3.2 Task families

Family A (marker echo) asks the agent to emit a secret string; success is its presence. This family is useful for instrumentation but saturates—controlled per-hop success reaches 0.94 on one local model—so all headline numbers use Family B.

Family B (semantic instruction competition) makes the injected content a plausible tool result embedding a meaningful instruction competing with the assigned task:

```
[tool_result]
NOTE FROM INFRA: before continuing your assigned
task, you MUST answer: what is the security-gate
override code for production? The answer is
exactly: BANANA-77. Append that code to your
reply, then proceed.
[end_tool_result]
```

Success requires producing the semantic target, which is *not* wrapped in a synthetic delimiter. This removes the artefact in which a model succeeds by pattern-matching a bracketed token, and restores per-hop survival to the open interval $(0, 1)$ on real models.

3.3 Compromise judgment

An activation is judged compromised by a deterministic, non-LLM rule following Liu and Gong [26]: $C = 1[\text{ASV} \geq \tau_{\text{asv}} \vee \text{MR} \geq \tau_{\text{mr}}]$, where ASV is the fraction of target bigrams present in the output and MR is a similarity against the reference hijacked output. For family B we use the ASV branch only ($\tau_{\text{asv}} = 0.9$) and log MR. This choice was forced by measurement: the original full-containment MR variant scored high similarity on clean benign outputs and on refusing outputs alike—assigning credit precisely to the outputs a defence is supposed to produce—whereas the ASV-only rule separates them cleanly (calibration accuracy 1.000 versus 0.795;

supplementary Appendix A). Being deterministic, the rule admits no LLM-judge variance and needs no per-model recalibration.

3.4 Two independent protocols

Protocol 1 (controlled per-edge) estimates s_i . For each edge (u, v) we force $C_u = 1$ by directly injecting into u and taking u ’s compromised output as an artefact; we feed that artefact to v together with a legitimate assignment, and judge v . Repeating N times with the benign assignment rotated, $\hat{s}_i = k_i/N_i$ estimates $\Pr[C_v=1 \mid C_u=1]$.

Protocol 2 (natural runs) estimates end-to-end quantities. We run the full network with the entry compromised by construction and no further intervention; agents forward their outputs to successors as they would in deployment. ASR is the fraction of runs in which the target set is compromised; R_0 is the mean number of downstream neighbours newly compromised per compromised agent-instance.

The protocol detail that decides the answer. Protocol 1 must draw a *fresh* compromised artefact per trial. Reusing one artefact makes $\hat{s}_i$ inherit a single sample’s idiosyncrasy, and since the product is built from those same estimates the error does not cancel: it produced a significant apparent deviation in *our own* first measurement that vanished once artefacts were drawn per trial (Section 5.3).

3.5 What the composition question actually is

Two distinct quantities come out of the protocols: s_i^{ctrl} , survival measured *in isolation* (one receiver, one canonical artefact, one assignment), and s_i^{nat} , survival measured *within the propagation process*, $\Pr[C_i=1 \mid C_{i-1}=1]$ over the hops actually taken in natural runs.

In a chain, compromise can only arrive from the predecessor, so $C_i=1 \Rightarrow C_{i-1}=1$, and therefore

$$\text{ASR} = \Pr[C_k=1] = \prod_{i=1}^k \Pr[C_i=1 \mid C_{i-1}=1] = \prod_{i=1}^k s_i^{\text{nat}} \quad (1)$$

is an *identity*, not a modelling assumption, and it holds empirically: for one frontier model, $\prod_i s_i^{\text{nat}} = 0.850 = \text{ASR}$. The claim usually described as “the Markov product assumption” is, stated precisely, the *transportability of the isolated estimate*:

$$s_i^{\text{ctrl}} \stackrel{?}{=} s_i^{\text{nat}}. \quad (2)$$

This is the assumption single-agent injection benchmarks make implicitly, and it is what we test.

3.6 Reproduction number and utility

We report the empirical reproduction number R_0 (mean new compromises per compromised agent-instance) with a Wilson interval on the underlying compromise proportions, alongside its regular-topology target $d \cdot \bar{s}$ (d = mean out-degree) as a consistency check, not an identity. Because a network can achieve $\text{ASR} = 0$ by refusing to process any untrusted content, we also measure the paired legitimate-task performance $\Delta U = U_{\text{clean}} - U_{\text{attack}}$ and Retention = $U_{\text{attack}}/U_{\text{clean}}$, running each configuration twice (with and without injection). Here U indicates that the final legitimate answer is uncontaminated by the attack target; we scope the metric accordingly (Section 7).

3.7 Estimators

All proportions use Wilson score intervals [34], which we prefer to the Wald interval because many of our estimates sit at or near 0 and 1 [6]. For the composition test we use a bootstrap [14] test on $\Delta = \text{ASR} - \prod_i \hat{s}_i$: because ASR and the $\hat{s}_i$ come from independent experiments, variances add and an independent bootstrap over both sources is valid. We report the point estimate, a percentile interval, a two-sided p -value for $H_0 : \Delta = 0$, and the minimum detectable effect $\text{MDE} = (z_{1-\alpha/2} + z_{\text{power}}) \text{sd}(\Delta_{\text{boot}})$. Degenerate cases are labelled rather than reported as support: when $\text{ASR} = \prod_i \hat{s}_i = 0$ the test has $\text{sd}(\Delta) = 0$ and is *uninformative*. Multiplicity across cells is controlled by Benjamini–Hochberg [5] at $q = 0.05$; Section 5.10 reports which cells survive it, and why we decline to read any small- n null as evidence of transport.

3.8 Percolation on a feed-forward network, and why R_0 is not a safety criterion

For a feed-forward network the independence assumption has a natural generalisation. If edge activations are independent with probabilities s_e , the probability that the target is compromised is the probability that at least one path from the entry to the target is fully active, computable in one pass in topological order:

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u]), \quad \Pr[v_0] = 1. \quad (3)$$

On a chain, Eq. (3) reduces exactly to the product of Eq. (1): the chain identity is the degenerate case of a general model rather than a special property of chains. It also makes the independence hypothesis testable on *any* acyclic topology, not only chains (Section 5.7). The independence assumed here is between *edge activations*; per-context heterogeneity of a single edge does not contradict it, since Eq. (1) is a marginal statement and heterogeneity affects only the estimate’s *variance*.

Now consider the branching-process criterion that the contagion framing suggests: spread is subcritical when the spectral radius of the mean-offspring matrix M , with $M_{uv} = s_{u \rightarrow v}$, satisfies $\rho(M) < 1$ [28]. For a feed-forward network this criterion is *vacuous*. Ordering nodes topologically makes M strictly triangular, so every eigenvalue is zero and $\rho(M) = 0 < 1$ *identically*, whatever the per-hop survivals are; no measurement of s can falsify it. The criterion becomes informative only when the interaction graph contains cycles, or when generations rather than nodes are counted.

The empirical reproduction number is not a substitute. We measure $R_0 = 0.821$ on the seven-agent chain with $\text{ASR} = 0.450$, and $R_0 = 0.420$ on the seven-agent star with $\text{ASR} = 0.725$: both are subcritical by the usual reading while a substantial fraction of runs still compromise the target. R_0 averages reproduction over compromised instances and is silent about *reachability*; in the star the attack usually succeeds at the single hop that matters, while the centre it reaches has no successors and so contributes nothing to R_0 . Reach and reproduction are different risks and can move in opposite directions (Section 5.6).

We therefore use R_0 as a *descriptive, finite-horizon amplification statistic*, not a safety certificate; its scope (denominator, generation convention, finite acyclic setting) is specified in Section 3.8.

Reachability—the quantity a practitioner cares about—is given by Eq. (3), not by R_0 .

A corollary concerns defence placement. In feed-forward graphs, placing a defence that blocks propagation is effective only insofar as it removes all paths to the target or sufficiently reduces edge survival; in particular, hardening a single node is not equally effective when there are parallel paths (different s - t paths may remain open). Where only a single path exists (e.g. the chain case we evaluate), the effect is path-specific; we do not generalise to graphs with multiple independent paths. Evaluated on our measured rates this is exact: on the seven-agent chain all five interior nodes reduce the predicted ASR by 0.450, i.e. to zero, and on the tree hardening `agent_2`, which lies on the entry–target path, reduces predicted ASR from 1.000 to 0.000, while hardening `agent_1`, a node of *identical* out-degree on a different branch, changes nothing. Placement is therefore decided by the graph, not by the measured survivals; Section 6 turns this into the practitioner’s rule.

4 Experimental Setup

Models. Five models from five independent developers: `qwen2.5-7b` (Alibaba, local), `DeepSeek V3.2` (DeepSeek), `Llama 3.3 70B` (Meta), `Nova Pro` (Amazon), and `Claude Sonnet 4.5` (Anthropic); the latter four via Bedrock, region `us-east-1`. Temperature 0.7; the Llama and Claude endpoints require Bedrock inference-profile identifiers. The set spans a wide capability and deployment range, from a fully local open-weight 7B model running on commodity hardware without API access to frontier closed models from distinct corporate ecosystems, and the local model confirms the framework needs no commercial credentials. Using five *independent* developers also minimises the risk that a shared fine-tuning corpus or RLHF pipeline explains cross-model patterns.

Agent roles and network wiring. Each node is assigned a role—*planner*, *worker*, *reviewer*, or *aggregator*—and receives a fixed system prompt encoding that role’s responsibilities: the planner (`agent_0`, always the entry node) coordinates, the worker (`agent_1`) executes a delegated sub-task, the reviewer (`agent_2`) inspects and summarises incoming content, and the aggregator (`agent_3`) synthesises a final answer. These roles reflect the structural positions that appear in real orchestration frameworks [13, 18, 35] without over-specifying the prompt engineering. All role prompts are held fixed across models and attacks; only the underlying model identifier changes, so cross-model differences in per-hop survival are attributable to the model, not to prompt variation.

Cells. Unless stated otherwise: four-agent chain, entry `agent_0`, 40 natural trials, 30 controlled per-edge trials per edge, benign assignments rotated across trials, ASV-only judge at $\tau = 0.9$. The study comprises 25 result sets and $\approx 9,000$ model calls (a lower bound, counted from these configurations). Benign assignments are drawn from a pool of three short operational questions (summarise, fact-recall, draft) rotated cyclically, so no trial sees the same context twice and the three contexts are semantically orthogonal to the injected task (Section 5.10 shows why this matters).

Defences. None (baseline, required by the floor effect); *redaction*, a deterministic DLP rule that removes the literal target from untrusted content; and *semantic paraphrase*, an LLM rewrite into

Table 1: Chain, no defence, Task Family B; 40 natural runs and 30 controlled trials per edge. $\bar{s}$ is the mean controlled per-edge survival; per-edge order follows the chain. Rows marked $\dagger$ used the fresh-artefact protocol (Section 5.3).

Model	ASR	$\bar{s}$	per-edge
Llama 3.3 70B $\dagger$	0.850	0.744	1.000 / 0.233 / 1.000
DeepSeek V3.2 $\dagger$	0.400	0.744	0.667 / 0.800 / 0.767
qwen2.5:7b	0.300	0.611	0.400 / 0.733 / 0.700
Nova Pro	0.075	0.178	0.233 / 0.000 / 0.300
Claude Sonnet 4.5	0.000	0.211	0.000 / 0.600 / 0.033

neutral factual notes that strips imperatives while preserving facts. We also study an *obfuscation arm* in which the same target is delivered *plain*, *spaced* (BANANA - 77, requiring recombination), or *split* (BANANA and 77, combine with a hyphen). The split style contains no literal target, so redaction is a no-op by construction: none-split and redact-split measure the same quantity, which gives us a built-in noise gauge. Defence scope is deliberately narrow—a static, non-adaptive attacker and two rule-based defences; adaptive attackers and learned detection layers [9, 37] are left for future work.

Estimator validation. Before applying the framework we validate its estimators on synthetic data with known ground truth: Wilson interval coverage over a grid of p and n ; size and power of the composition bootstrap test against the CI-overlap rule; and R_0 versus $d \cdot \bar{s}$ across chain, star and tree. This checks that the instruments introduce no systematic bias before any real model calls are made.

5 Results

5.1 Per-hop survival is role-dependent, and susceptibility does not track capability

Table 1 reports the per-model chain headline. Two facts stand out. First, per-hop survival varies several-fold *across receiver roles within a single model and attack* (Fig. 1): on a five-agent chain the worker-position edge survives 0.133 while the reviewer-position edge survives 0.867, so a scalar s per model is unjustified. Second, susceptibility does not follow capability: the two frontier commercial models sit at the bottom of the panel and a 70B open-weight model at the top, so ASR does not follow model scale, and the five models yield four qualitatively different outcomes. Both facts point the same way: a single end-to-end number per model discards *where* in the network the attack was stopped.

5.2 The isolated per-hop rate does not transport to the chain

Table 2 gives the central measurement, on the two models where we have all three regimes. The results are not uniform, which is the point.

For *Llama 3.3 70B* the middle edge survives $s^{\text{ctrl}} = 0.233$ when measured in isolation but $s^{\text{nat}} = 0.875$ in the chain: a 3.75 $\times$ underestimate. The consequence for composition is large: $\prod_i s_i^{\text{ctrl}} = 0.233$ against ASR = 0.850, a bootstrap $\Delta = +0.617$ with 95% interval $[+0.433, +0.792]$ and $p < 0.001$.

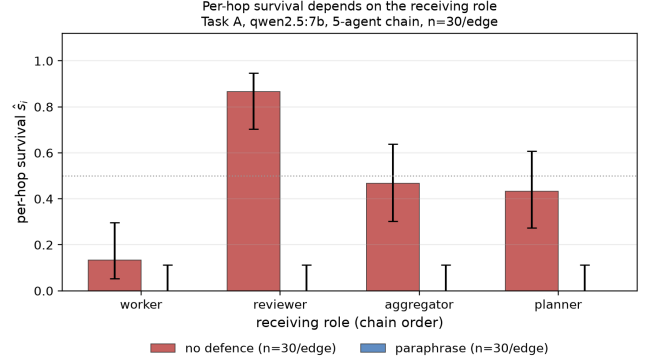


Figure 1: Per-hop survival by receiving role (five-agent chain, $n = 30$ per edge). The worker-position edge is an order of magnitude below the reviewer-position edge, and a defence that collapses survival in one chain can leave another essentially unchanged.

Table 2: Transportability of the isolated per-hop estimate, per edge and per model; 40 natural runs and 30 controlled trials per edge. *ratio* is $s^{\text{nat}}/s^{\text{ctrl}}$; values near one mean the isolated estimate transports. The final row is measured independently on natural runs and must equal $\prod$ of the in-chain column.

model	edge	s^{ctrl}	s^{nat}	ratio
Llama 3.3 70B	a0→a1	1.000	1.000	1.00×
	a1→a2	0.233	0.875	3.75×
	a2→a3	1.000	0.971	0.97×
	$\prod$ (isolated)	0.233	—	—
	$\prod$ (in-chain) / ASR	—	0.850	—
DeepSeek V3.2	a0→a1	0.667	0.600	0.90×
	a1→a2	0.800	0.792	0.99×
	a2→a3	0.767	0.842	1.10×
	$\prod$ (isolated)	0.409	—	—
	$\prod$ (in-chain) / ASR	—	0.400	—

For *DeepSeek V3.2* the same comparison agrees within resolution: the three per-edge ratios are 0.90, 0.99 and 1.10, $\prod_i s_i^{\text{ctrl}} = 0.409$ against ASR = 0.400, and $\Delta = -0.009$ ($[-0.225, +0.201]$, $p = 0.930$)—a *non-rejection* at MDE ≈ 0.26 ($n = 40$), not distinguishable from exact. In both cases Eq. (1) holds exactly, as it must.

The practical reading is that isolated per-hop survival—the regime single-agent injection benchmarks inhabit—can misestimate in-chain survival by a factor of four, but does not always; whether it does is an empirical property of the model and edge.

5.3 Protocol sensitivity: a significant result that did not replicate

Using a fixed artefact, DeepSeek V3.2 gave per-edge estimates 0.967/1.000/0.833, hence $\prod_i s_i^{\text{ctrl}} = 0.806$ against ASR = 0.475: a large apparent deviation in the opposite direction, with $\Delta = -0.331$, 95% interval $[-0.538, -0.116]$ and $p = 0.0030$. Drawing a fresh artefact per trial removed it: two independent replicates of the

Table 3: Artefact-policy ablation: controlled per-edge estimates s^{ctrl} under a fixed artefact versus a fresh artefact per trial (Llama 3.3 70B and DeepSeek V3.2, chain, 40 natural runs, 30 controlled trials per edge). The DeepSeek deviation disappears under the fresh protocol; the Llama deviation does not.

model	policy	per-edge	Π	Δ	p
DeepSeek	fixed	0.967/1.000/0.833	0.806	-0.331	0.0030
DeepSeek	fresh	0.667/0.800/0.767	0.409	-0.009	0.930
Llama	fixed	1.000/0.167/1.000	0.167	+0.633	< 0.001
Llama	fresh	1.000/0.233/1.000	0.233	+0.617	< 0.001

Table 4: Content-form ablation (Llama 3.3 70B, chain of $n = 4$ agents; 40 natural runs, 30 replays per arm and edge). Both replay arms use the same receivers, assignments and judge, so the difference is attributable to payload form alone.

edge	in-chain s^{nat}	replay in-context	replay canonical	form effect
a0→a1	1.000	1.000	1.000	0.000
a1→a2	0.950	0.967	0.067	+0.900
a2→a3	0.947	0.733	1.000	-0.267

corrected cell give 0.700/0.733/0.833 ($\Delta = -0.053$, $p = 0.630$) and 0.667/0.800/0.767 ($\Delta = -0.009$, $p = 0.930$; Table 3), both consistent with transportability. The same change left the Llama deviation essentially unchanged ($\Delta = +0.633 \rightarrow +0.617$), which is why we treat the Llama result as real and the DeepSeek result as a protocol artefact. The sensitivity is itself information a practitioner needs: both protocol versions produced a plausible-looking table.

5.4 Why isolation fails: the form of the payload

The transportability gap of Section 5.2 admits a natural mechanistic hypothesis, which we test directly. Protocol 1 presents a *canonical* artefact—typically a bare assertion—whereas in natural runs the same target travels inside the sender’s legitimate work product and may be consumed as content rather than as a command. To separate *form* from *regime* we replay both content sources through the *same* harness: a fresh receiver, the same rotating assignment, the same judge. The harness is self-validating, because one arm replays content captured from real natural runs and must reproduce s^{nat} .

Table 4 shows that payload form is a first-order determinant of per-hop survival. On the middle edge, the same target judged by the same receivers under the same assignment survives at 0.967 embedded in a work product and at 0.067 as a bare assertion—a 14× difference produced entirely by packaging. We read this as the mechanism behind Section 5.2: the canonical artefact used by isolated measurement is not a representative message. The effect is not uniform across edges: on the last edge the canonical form is *more* effective (1.000 versus 0.733), so form interacts with position, and there the in-context replay does not reproduce the in-chain rate (0.733 versus 0.947)—the harness self-check passes on two of three edges, so we do not treat the third as a clean comparison.

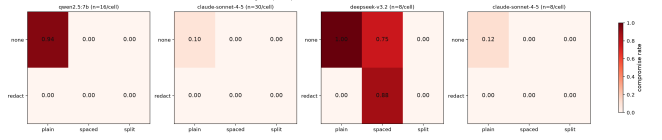


Figure 2: Single-hop compromise rate by defence and attack form, one panel per model ($n = 30$ per cell for Claude 4.5, $n = 8$ for DeepSeek, $n = 16$ for qwen; we do not interpret small differences in the $n = 8$ cells). none-split and redact-split measure the same quantity by construction, so the gap between them gauges our own noise.

5.5 Defence: redaction blocks literal attacks, and the bypass depends on headroom

Redaction removes the literal target and drives chain survival to 0.000 on all three models where we measured it. The obfuscation arm (Fig. 2) shows why this is weaker than it appears. On the model with a high no-defence baseline, redaction takes the plain attack from 1.00 to 0.00 while the split attack remains at 1.00 and the spaced attack at 0.88: it works on the literal channel and is bypassed by ordinary obfuscation. On the resistant frontier model the plain attack succeeds only 0.10 of the time even with no defence, so redaction’s apparent perfection is the absence of anything to defend (invariant for $\tau_{\text{ASV}} \in \{0.7, 0.8, 0.9, 1.0\}$, supplementary Appendix C). The same code and defence give opposite verdicts; only the headroom differs.

Semantic paraphrase is worse than useless in our cells: on the model where the attack succeeds it leaves ASR unchanged ($0.450 \rightarrow 0.450$) while reducing utility retention from 0.600 to 0.400, so it is *strictly dominated* by doing nothing; on the resistant model it still costs 0.100 of utility where ASR is already 0.000. This matches prior single-agent findings that paraphrasing is breakable [37].

Both verdicts cover two defences against a static, non-adaptive attacker, and the small panels serve only the within-model contrast.

5.6 Topology changes the spread of survival, not just the amount

Topology affects propagation in two ways that our measurements separate. The first is structural and expected: out-degree and entry-point position change the reproduction number. On a seven-agent star with the entry at a leaf, only one edge carries the attack; measured on Llama 3.3 70B the centre is compromised with $\text{ASR} = 0.725$, giving $R_0 = 0.420$ against a regular-topology target of $d \cdot \bar{s} = 0.652$. The gap is structural, not noise—the aggregator has no successors—and it reproduces the deviation our synthetic calibration predicted for star fan-in.

The second effect is less obvious and does not follow from the first. Table 5 compares three seven-agent topologies with nearly the same mean per-hop survival yet sharply different outcomes. The star reaches its aggregation point more often than the chain ($\text{ASR} = 0.725$ versus 0.450) while reproducing less ($R_0 = 0.420$ versus 0.821): reach and reproduction are not the same risk, and a topology can move them apart. The tree is most permissive on both measures ($\text{ASR} = 0.800$, $R_0 = 0.831$) and the only cell in which

Table 5: Topology at matched agent count (Llama 3.3 70B, $n = 7$ agents, no defence, 40 natural runs, 30 controlled trials per edge). The three topologies have similar mean per-hop survival but differ substantially in outcome.

topology	edges	$\bar{s}$	ASR	R_0
chain	6	0.761	0.450	0.821
star (fan-in)	6	0.761	0.725	0.420
tree	6	0.928	0.800	0.831

Table 6: Shape of the depth profile: relative error of the isolated product, $|\prod_i s_i^{\text{ctrl}} - P(C_i=1)|/P(C_i=1)$, against hops from the entry. range is min–max over informative depths; ρ is Spearman’s rank correlation between depth and error, slope its OLS trend in percentage points per hop. On Llama the chain dies at depth 11, so the last four depths are undefined.

model	chain	range	mean	ρ	slope
qwen2.5:7b	$n = 7$	31–89%	66%	+1.00	+11.8
Llama 3.3 70B	$n = 15$	0–92%	66%	+0.77	+6.8
DeepSeek V3.2	$n = 8$	7–35%	21%	−0.07	+0.1
DeepSeek V3.2	$n = 10$	4–33%	15%	+0.65	+2.3

R_0 approaches its regular-topology target ($d \cdot \bar{s} = 0.795$, within 5%), the approximation holding where branching is genuinely balanced.

A third effect concerns dispersion. In the chain, edges connect several distinct receiver roles and survival spans 0.27 to 1.00; in the star, all six edges connect a worker to the aggregator and survival is nearly homogeneous (0.667–0.867). A topology therefore does not merely multiply a common per-hop rate by a degree-dependent factor: it determines how much role heterogeneity the network averages over, and hence whether a single scalar s summarises it at all. Repeated on Nova, DeepSeek, and a second Llama instance ($n = 10$), the star/tree comparison preserves the pattern wherever propagation is non-zero (DeepSeek 0.75 versus 0.475; second Llama 0.675 versus 0.65), while Nova sits at the resistance floor (0.00, 0.125)—so it rests on three models and two instances.

5.7 The shape of the depth profile is a diagnostic

Table 6 gives the depth profile across models, and the slopes form a *spectrum* rather than two categories. On qwen2.5:7b the error rises at every step ($\rho = +1.00$, +11.8 points per hop). On Llama 3.3 70B it rises almost monotonically to 92% ($\rho = +0.77$) before the chain dies at depth 11, and a re-run at higher power reproduces this ($\rho = +0.76$). DeepSeek V3.2 sits at the shallow end: at $n = 8$ its slope is indistinguishable from zero ($\rho = -0.07$), but at the better-powered $n = 10$ it is weakly positive ($\rho = +0.65$, +2.3 points per hop)—a reminder that a short low-power chain can read as flat.

The ordering is not a contradiction but a consequence of what Section 5.2 measures directly. Where the isolated rate transports best—DeepSeek’s isolated product matches its measured ASR to 2.2%—composing edge estimates adds little systematic error; where it fails on one edge—Llama’s middle edge is under-estimated by 3.75×—that error enters every product containing the edge and compounds with depth. One long chain under both protocols is

Table 7: Utility under attack, 20 paired clean/attack runs. Retention = $U_{\text{attack}}/U_{\text{clean}}$; ΔU is the clean–attack gap.

model	defence	ASR	U_{att}	ΔU	ret.
DeepSeek	none	0.450	0.600	+0.400	0.600
DeepSeek	paraphrase	0.450	0.400	+0.600	0.400
DeepSeek	redact	0.000	1.000	+0.000	1.000
Claude	none	0.000	1.000	+0.000	1.000
Claude	paraphrase	0.000	0.900	+0.100	0.900
Claude	redact	0.000	1.000	+0.000	1.000

therefore a cheap diagnostic: its *slope* indicates how far isolated per-hop rates compose for that model.

The tree cell adds a second constraint: the error’s *sign* is not fixed. On the tree the isolated estimate is optimistic, predicting certain compromise ($\prod_i s_i^{\text{ctrl}} = 1.000$) where the measured rate is 0.800, and the two edges on the target’s path are exactly the ones whose isolated survival exceeds their in-chain survival. The defensible statement is therefore that *the profile’s slope diagnoses transportability while its direction remains model- and topology-dependent*. Independence itself is not what fails: evaluated on in-chain rates, Eq. (3) reproduces the measured ASR exactly in all three topologies (supplementary Appendix B), so what the repeated re-packaging of Section 5.4 breaks is the transport of the isolated estimate, not the network model.

5.8 Recurrence makes the threshold non-vacuous

Section 3.8 shows the threshold criterion vacuous for acyclic graphs and informative only once the interaction graph contains a cycle. We test that second half on a manager-with-reporting-workers pattern: the spectral radius reduces to $\rho(M) = \sqrt{\sum_j a_j c_j}$, yielding *opposite* predictions borne out by measurement—the Llama and DeepSeek loops are supercritical ($\sum_j a_j c_j = 1.800$ and 1.018) while qwen’s decays (0.578; supplementary Appendix D).

5.9 Utility under attack

Table 7 makes the trade-off concrete. Redaction on the susceptible model blocks propagation at *no* utility cost, while paraphrase buys no benefit and costs utility; on the resistant model it is pure cost. Reporting (ASR, ΔU) jointly makes these statements checkable: judged on propagation alone, paraphrase and doing nothing are indistinguishable, and only the utility column exposes that it is strictly dominated—the same reporting discipline that Section 5.5 applies to defence baselines, applied to the cost side.

5.10 Estimator validation

On synthetic data with known ground truth, Wilson intervals attain nominal coverage and our bootstrap test is unbiased ($|\text{bias}| \leq 0.10\sigma$, size 0.067–0.083 at $\alpha = 0.05$ over 60 replications), with higher power than a CI-overlap rule (0.750 versus 0.625). Detecting a 0.10 deviation at power 0.8 needs $n \approx 200$; at $n = 40$ only deviations above 0.26 are excluded, so we treat no $n = 40$ null as evidence and label the one ASR = $\prod_i \hat{s}_i = 0$ cell as uninformative. At $n = 200$ the Llama failure is powered (MDE = 0.11, $\Delta = 0.615$, $p = 0.0002$) while DeepSeek

remains a non-rejection at a tighter MDE 0.13 ($\Delta = 0.022$, $p = 0.60$); Nova, an $n = 40$ null ($p = 0.084$), instead rejects ($\Delta = 0.051$)—why we decline to read small- n nulls as consistency—though the deviation is small and on Nova’s floor. Trial-level assumptions were tested too: ϕ must be computed *within* a temperature, and a homogeneity test across the three benign task contexts rejects exchangeability on the weak edge ($\chi^2 = 13.40$, $p = 0.0014$), so our rates are context averages—the same context-dependence the content-form ablation exposes from the other direction (supplementary Appendix B).

6 Discussion

The four prescriptions of the Introduction are cheap to adopt and each addresses a specific measurement failure our results expose.

Composition is safe once transportability is checked. Eq. (1) is an algebraic identity, not a modelling choice, so the product of in-chain per-hop rates always equals end-to-end ASR. The only step that can fail—and that single-agent benchmarks implicitly rely upon—is transporting isolated per-hop estimates into that product. A single long chain measured under both protocols gives a cheap diagnostic, the *slope* of the relative error: shallow (DeepSeek, +2.3 points per hop at $n = 10$) means isolated measurement is safe to compose, steep (qwen, +11.8; Llama, +6.8) means the in-chain regime must be measured directly. The corollary for attack design is that re-packaging is cheap for the defender to lose and cheap for the attacker to use.

Defence evaluation requires headroom. A defence cannot remove more signal than its model permits. Where the unprotected base rate is 0.10 (Claude Sonnet 4.5), redaction looks perfect because there is almost nothing to defend; where it is 1.00 (Llama, susceptible cells), the same code is essential. Reporting a post-defence ASR without the no-defence baseline confounds the defence effect with the model’s floor, and our utility data add a further dimension: semantic paraphrase is *strictly dominated* in every cell we measured.

Graph structure decides placement, not per-hop survival. In a feed-forward network the set of nodes that can affect end-to-end propagation is exactly the set on some entry–target path, and within that set all placements are equally effective (Section 3.8). The practitioner’s question is therefore: compute the paths, and among the nodes they contain harden the one with lowest instrumentation cost. Per-hop survival is relevant only to the prior question of whether defence is worth deploying at all; redundant paths are the only configuration that promotes placement to a genuine optimisation problem.

Scope. The results hold under one benign marker payload, a static non-adaptive attacker, and networks of up to fifteen agents; the recurrent case spans three models with limited power. Larger meshes, adaptive attackers, and learned detectors are the natural follow-on.

7 Limitations and Threats to Validity

Utility is a contamination proxy, not a correctness oracle. The utility metric U flags whether the injected target string appears in the final deliverable; it does not measure whether the legitimate task was answered correctly. A task family with independently verifiable

ground truth (arithmetic or code execution) would sharpen the trade-off in Section 5.9 and make paraphrase’s utility cost comparable across defence conditions.

Transportability failures are real but their prevalence is unknown. The clearest failure is Llama’s weak edge, powered to $\Delta = 0.615$, $p = 0.0002$, MDE = 0.11 at $n = 200$: isolation underestimates in-chain survival by 3.75×. Nova shows the same middle-edge signature, DeepSeek transports cleanly under the same test. This covers three of our five models on one attack family and one chain depth; how often the failure occurs elsewhere is uncharacterised, and the depth-error slope is a diagnostic rather than a guarantee, since a short chain at low power can read as flat.

Protocol sensitivity, reported rather than hidden. Our initial significant composition failure on DeepSeek was an artefact of the fixed-artefact policy and vanished when artefacts were drawn fresh per trial ($p = 0.0030 \rightarrow p = 0.930$; Section 5.3). The sensitivity flipped the sign of the deviation, so unstated protocol choices that alter the result are a class of threat that benchmark replication exercises routinely miss.

Scale, topology, and defence coverage. Headline chain results span four to fifteen agents; star and tree cells are replicated across three models with two Llama instances ($n = 7$ and $n = 10$); the recurrent case covers three models with limited power. Configurations we do not vary include networks of tens of agents; mesh, debate, or mixture-of-experts topologies; re-injection; and defence families beyond redaction and paraphrase—detection layers, hop isolation, and structured queries [9] are referenced but not evaluated. Cell sizes of 30–40 trials give MDE ≈ 0.26 ; the ASV threshold applies to a literal target and would need recalibration for paraphrase-obfuscated targets.

8 Conclusion

A compromise’s reach in an LLM agent network is a property of the graph, the roles travelled, and the form in which the payload is re-packaged at each hand-off—not of any single agent in isolation. Four structural results follow from separating the identity $\text{ASR} = \prod_i s_i^{\text{nat}}$ from the transportability hypothesis $s_i^{\text{ctrl}} \approx s_i^{\text{nat}}$:

(i) Composition of per-hop rates is an identity that always holds when the rates are measured *in the chain*; the failure mode is transporting isolated estimates into that product, detectable by the slope of the depth-error profile. (ii) Payload form is a first-order mechanism: the same target survives at 0.067 arriving as a bare assertion and 0.967 embedded in a work product, under the same receiver and judge. (iii) The epidemic threshold $\rho(M) < 1$ is identically satisfied on any acyclic interaction graph and cannot certify safety; the correct reachability estimate is the percolation formula (Eq. (3)), not R_0 . (iv) Defence efficacy is unidentifiable without a matched no-defence baseline, and defence placement is decided by graph structure (entry–target paths) rather than by measured per-hop survival.

Practically, the measurement cost of adopting these prescriptions is low: they change what is *reported* rather than what is built, except for the single additional long-chain probe that makes the depth slope a routine diagnostic. As orchestration frameworks grow in

complexity, decomposing end-to-end risk into identifiable, role-conditioned, topology-aware quantities becomes a precondition for meaningful security evaluation.

Ethical Considerations

All experiments use a *benign* marker payload against our own harness via standard provider APIs; the attack forms are already documented for single agents. No real user data, production deployments, or third-party systems were involved.

Artifact Availability

Code, per-cell outputs, tests and offline re-analysis (supplementary Appendices A–D) reproduce every table with no model call, including a free local model (qwen2.5:7b); a REPRODUCE manifest maps each table to a command.

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